

Lab: Branching and Merging in Git

Objective

- Understand branching and merging.
- Create a new branch and merge it with the master branch.

Commands – Branching

1. Create a new branch GitNewBranch

`git branch GitNewBranch`

2. List all local and remote branches

`git branch -a`

3. Switch to the new branch

`git checkout GitNewBranch`

4. Verify the status of the repository

`git status`

5. Commit the changes

`git add .`

`git commit -m "Added file1.txt in GitNewBranch"`

`git status`

Commands – Merging

1. Switch to the master branch

git checkout master

2. View differences between master and branch

git diff master GitNewBranch

3. View visual differences using P4Merge

git difftool master GitNewBranch

4. Merge the branch into master

git merge GitNewBranch

5. View commit history after merging

git log --oneline --graph --decorate

6. Delete the branch and check status

git branch -d GitNewBranch

git status